



Competitive Programming Camp

CPCCU AI-Powered Web App Development & Deployment Hackathon 2026

Title

Build a Smart Digital Campus Hub for City University - CampusOS for City University

Background

City University is a busy, growing campus with more than 10+ active clubs - cultural, technical, debate, sports, and academic - alongside multiple departments, research labs, a shuttle bus network serving several routes, and administrative offices that all generate information students need on a daily basis: exam schedules, class cancellations, contest announcements, form deadlines, seminar notices, lab classes, and more.

None of this information lives in one place. Instead, it is fragmented across:

- Facebook groups** - one per club or department, each with its own posting habits, admins, and inconsistent update frequency

- Messenger group chats** - where important announcements get buried within minutes under unrelated chatter, memes, and side conversations

- Google Forms** - used ad hoc for event registration, feedback, or complaints, with no central place listing which forms currently exist or where to find them

- Word of mouth and physical notice boards** - which only reach students who happen to be in the right place at the right time

A new student joining City University has no map of any of this. They don't know which Facebook groups to join, which Messenger groups matter, or which forms are currently open. Even senior students who have learned to navigate the chaos still waste time re-checking five different channels for one piece of information, and inevitably miss things that were only posted in a channel they don't follow closely.

The result is a campus where useful, time-sensitive information exists - it's just scattered, unsearchable, and disconnected, and the cost of that fragmentation falls hardest on the students who can least afford to miss it: newcomers still learning how the university works.

The Problem, in Detail

Walk through an ordinary week for a CU student and the friction shows up constantly:

- Missed events** - A student wants to attend a workshop or contest hosted by a club, but only finds out about it a day late because it was posted once in a Facebook group they don't check regularly. There is no reminder, no unified calendar, and no way to browse "everything happening this week across all clubs" in one view.

- Bus confusion** - A student misses their shuttle because the route and timing were shared once in a Messenger group months ago and are now buried under hundreds of unrelated messages. There's no dedicated, always-current place to just look up "when is the next bus on my route."

- Class cancellations** - A class gets cancelled or moved at the last minute, but the notice only reaches the subset of students who happen to be active in the right group chat at that exact moment. Others show up to an empty room.

- Lost academic resources** - A student needs last semester's question papers, a specific lecture's notes, or a lab manual, and ends up manually asking around in multiple group chats instead of searching a shared, organized archive. Useful material that other students already have sits scattered across personal drives and old chat threads instead of being reusable by everyone.

- Lost items with no system** - A student loses their ID card, a bag, or a calculator somewhere on campus. Their only option is a single Facebook post that scrolls out of view within hours, with no

structured way for someone who found the item to connect it back to the right owner, and no way to browse "what's been found recently."

No single starting point - Perhaps the deepest issue: there is no one URL, one app, or one place a student - especially a first-year student - can go to and reasonably expect to find what they need.

Every new student effectively has to rediscover the scattered ecosystem for themselves.

The common thread across every one of these situations is the same: **City University has no single source of truth for campus life**, and that gap quietly costs every student time, missed opportunities, and avoidable stress - a cost that compounds for anyone new to campus who hasn't yet mapped out the informal channels everyone else relies on.

The Challenge

Design and build **CampusOS - City University**, a unified web application that becomes that missing single source of truth for the campus.

The core objective is to replace the scattered, single-purpose, largely unsearchable channels currently in use (Facebook groups, Messenger groups, ad hoc Google Forms) with one structured, reliably updated, browsable platform, designed specifically around how a real CU student - new or returning - actually moves through their day: checking what's happening, catching the bus, keeping up with class changes, finding study material, and recovering lost belongings.

The platform should feel like it was built for City University specifically, not like a generic template.

Wherever possible, ground your design decisions in the concrete scenarios described above - a good submission should be able to point to a specific piece of UI or a specific feature and say exactly which daily friction it removes.

AI features are encouraged but explicitly optional. The priority for judging is a genuinely usable, reliable, functional platform first. If you choose to add AI, it should make an existing module meaningfully faster or easier to use (smarter search, automatic summaries, smarter recommendations) - not be bolted on purely to tick a box. A polished, fully working platform with no AI will be judged more favorably than a shaky platform with AI features stapled onto it that don't reliably work.

Core Modules (build at least 2)

You must choose and fully build **at least two** of the four modules below. "Fully built" means the module works end-to-end with real data flowing through it - not a static UI mockup with placeholder content. For each module you choose, think through the full loop: how does content get in, how is it organized, and how does a student actually find and use it.

1. Club & Event Engine

A single, unified feed covering all club and department events across campus, replacing the need to check 20+ separate Facebook groups.

- A browsable list/feed of upcoming events across all clubs, not just one club's events

- Each event should have essential details: what it is, which club is hosting it, date/time, and location

- A way for students to register or RSVP for an event

- A way to check attendees in at the event itself (e.g. a QR code generated per registration, scanned at the door)

- Consider how a student filters or searches this feed - by club, by date, or by type of event

2. Resource Hub

A shared, organized, searchable archive for notes, past question papers, and notices - replacing the scattered personal-drive-and-group-chat approach students currently rely on.

A way for students (or admins) to upload material: notes, question papers, official notices
Material should be organized in a way that's actually browsable - by course, department, or category, not one giant unsorted list
A working search over this material, so a student can find "previous questions for [course]" without manually scrolling
AI-powered semantic search is a strong optional addition here, but a well-built keyword/category search fully satisfies the core requirement

3. Smart Helpdesk

A single, reliable place for students to get answers about CU rules, the bus schedule, and exam-related info - replacing the "ask in the group chat and hope someone answers" pattern.

A structured, well-organized set of information covering the kinds of questions students actually ask: bus routes and timings, key university rules, exam-period logistics

This can be built as a clean, well-organized FAQ/info browsing system, or as an AI chatbot if you choose to build one - either approach satisfies the module, as long as it's genuinely useful and easy to search

If you do build a chatbot, make sure its answers are grounded in real CU information you've provided it, not generic filler

4. Lost & Found / Complaint Box

A structured system for reporting and recovering lost items or filing complaints - replacing the single scrolling Facebook post that disappears within hours.

A way to post a lost item (or a found item) with a description, photo, and location/date

A way to browse currently posted lost/found items, so someone who found something can check if it's already been reported, and someone who lost something can check if it's turned up

A way for a student to file a general complaint and track that it was received

AI image matching between "lost" and "found" posts is an optional enhancement - a clean, searchable, well-organized post-and-browse system meets the core requirement on its own

Must-Haves for Judging

These are the non-negotiable requirements your submission will be checked against, regardless of which modules or AI features you choose to build:

Live deployed link with working authentication The application must be reachable at a public URL during the judging period, and students must be able to sign up and log in. If judges can't access or log into your app, they can't evaluate it - deployment and auth are treated as foundational, not optional polish.

Documentation A clear, well-written Markdown documentation file (e.g. README.md) that a judge - or a new developer - could read and immediately understand:

What the project does and what problem it solves

Which modules you built and how each one works

The technology stack used

How to run the project locally (setup/install/run instructions)

Any environment variables, seed data, or demo credentials needed to test it

At least the core modules functional Whichever two (or more) modules you chose to build must actually work end-to-end when a judge tries them - not just look good in a screenshot. A feature that's visually complete but doesn't function (e.g. a "register" button that does nothing, a search box that returns nothing) does not satisfy this requirement.

Real-world usability description for a City University student Your documentation must include a clear, concrete section explaining exactly how this platform solves the daily friction described above for a real CU student. Don't describe it abstractly - walk through specific scenarios (e.g. "a first-year student checking the Resource Hub the night before an exam instead of messaging five different groups") and explain how your build addresses each one.

Presentation & Demo Video Per the official rulebook, every team must submit a Presentation & Demo Video alongside the written submission - there is no separate live presentation unless the organizers announce one. The video should walk through: **Problem** → **Solution** → **Demo** → **Features** → **Technical Implementation**. Screen recording with voice-over, slides, a live walkthrough, or a mix of these are all acceptable. Upload it to Google Drive or another accessible cloud storage service and submit a public/viewer link.

No exposed credentials Double-check your repository and public submission before sending it in - never commit private API keys, passwords, or other sensitive credentials. This is called out explicitly in the official rulebook.

AI features - a chatbot, semantic search, auto-summarization, image matching, or anything else - remain a **bonus**, not a requirement, and per the official rulebook, **AI usage itself is not a judging criterion**. They are not part of this must-have list.

Evaluation Principles (from the official rulebook)

Keep these in mind while you build - they describe how judges weigh trade-offs, not separate marks:

Working product > Concept only

Relevant solution > Unnecessary complexity

Effective implementation > Technology for the sake of technology

Your scope will be judged **in the context of the 24-hour challenge** - a smaller, fully working build beats an ambitious, half-finished one.

Bonus Ideas (optional)

AI notice summarizer - image to text

AI timetable conflict checker

AI-based club recommendation for freshers

AI semantic search across uploaded resources

AI chatbot for the Smart Helpdesk

Project Submission

Join the CPCCU GitHub organization - request/accept the invite so your team has access to the org.

Create your team's repository inside the CPCCU GitHub org - do not submit from a personal/outside repo; the project repo must live in the org.

Record and upload your Presentation & Demo Video - covering Problem → Solution → Demo → Features → Technical Implementation - to Google Drive or another accessible cloud storage service, and grab a public/viewer link.

Fill out the official submission form before the deadline, including:

Team name and members

Project name and short description

Problem & proposed solution

Major features

Live deployment link
Source code repository link (inside the CPCCU org)
Demo credentials (if login is required)
Technology stack (major languages, frameworks, databases, APIs, AI tools, etc.)
Presentation & Demo Video link
Link to your documentation

Before submitting, double-check that your deployment, repository, and video links are all actually accessible
- broken links can't be judged.

Conclusion

Flow	How CampusOS Hits It
Problem	Scattered clubs/notices/buses across FB, Messenger, Forms - no single source of truth
Solution	One unified, functional hub built around real CU student pain points
Implementation	2+ working core modules, live auth, clear documentation
Real-World Usability	Solves the exact daily friction described by CU students

Official Judging Criteria (100 marks)

Criteria	Marks
Problem Understanding & Relevance	15
Innovation & Creativity	20
Functionality & Completeness	25
UI/UX & Accessibility	15
Technical Implementation	15
Presentation & Demo	10